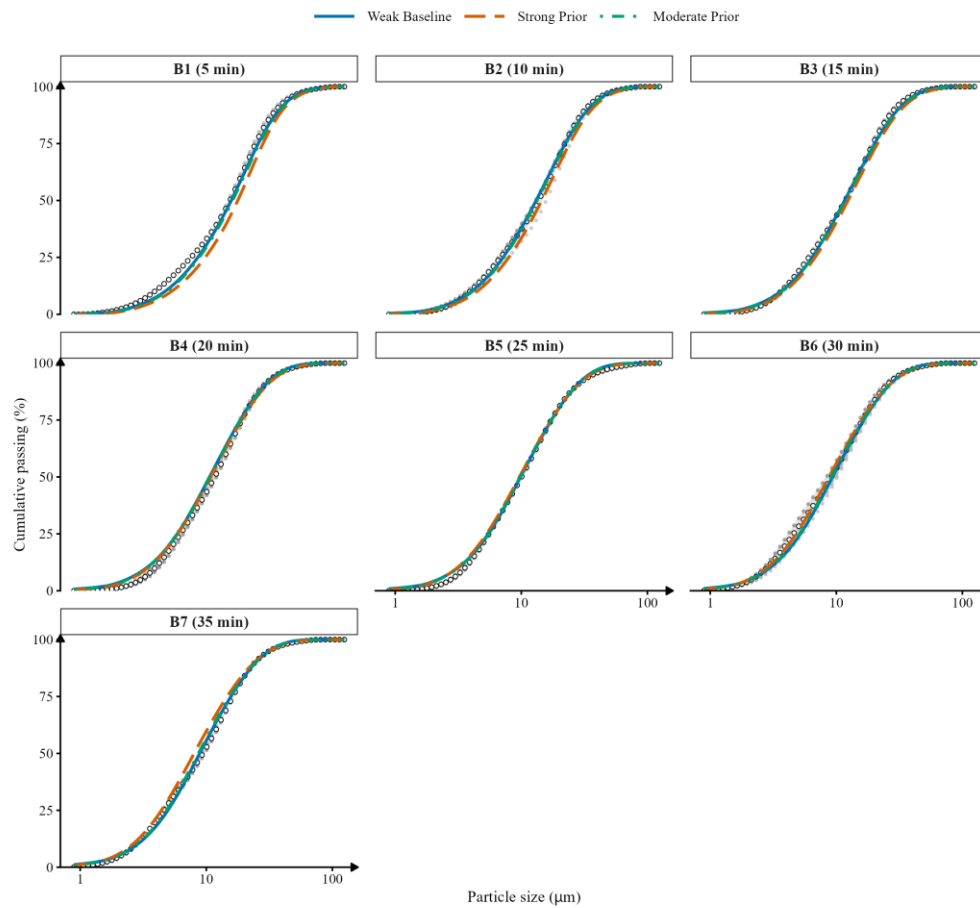


# Supplementary Materials for Mechanistic Investigation of Particle Size Evolution during Ultrafine Ball Milling of Ground Granulated Blast Furnace Slag Using a Nonlinear Population Balance Model

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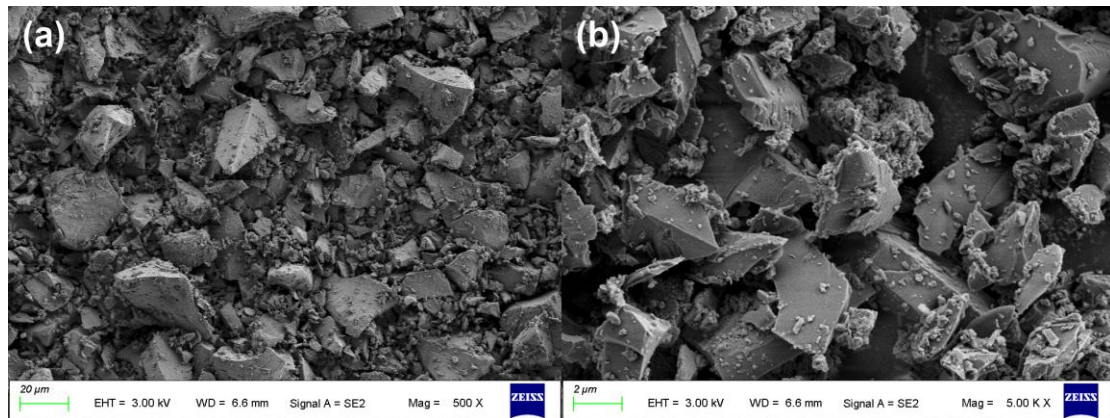
# Supplementary figures

## Bayesian prior-sensitivity analysis for all milled samples

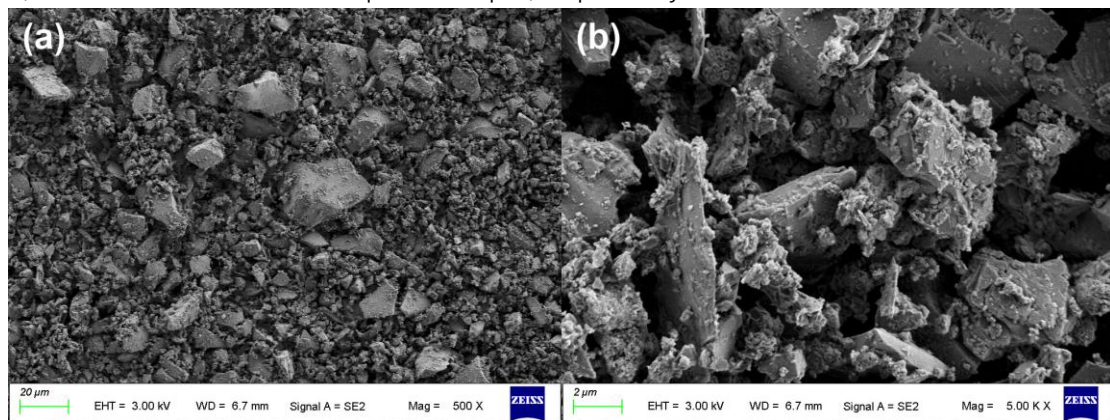


**Figure S1.** Observed cumulative particle-size distributions (PSDs) for B1–B7 and posterior-median predictions under the three prior scenarios. Grey points show the six experimental replicates, white-filled points and grey bands show their pointwise mean  $\pm 1$  standard deviation, and colored lines show the mean of the six replicate-specific posterior-median fitted curves.

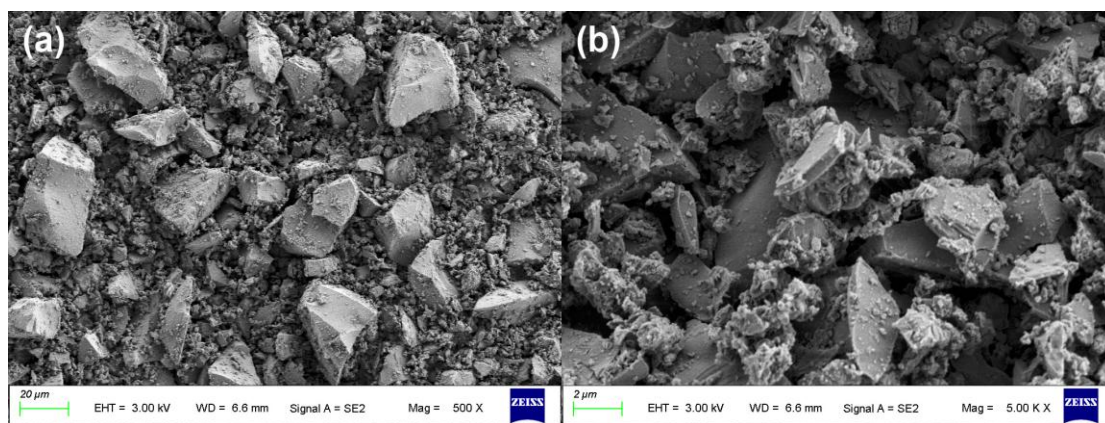
## Scanning electron microscopy of all samples



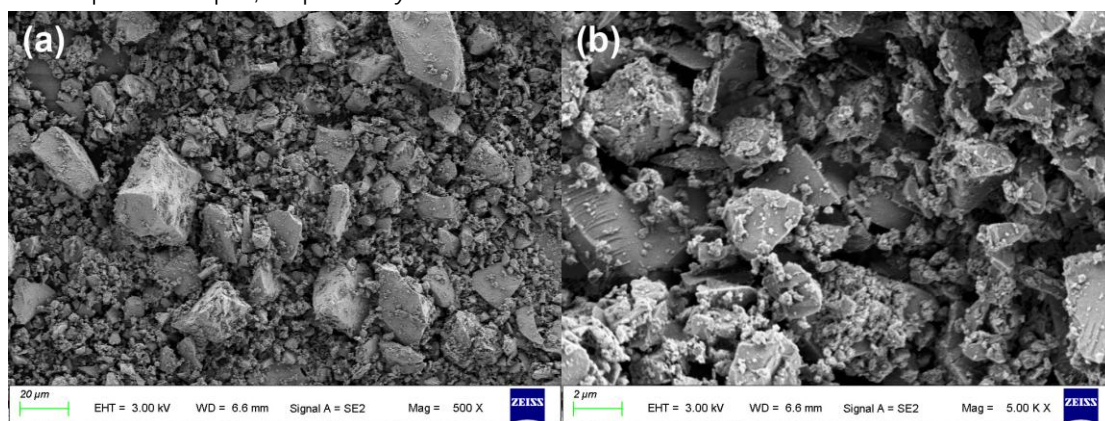
**Figure S2.** SEM micrographs of the as-received GGBS sample B0 (0 min). Left: 500 $\times$ ; right: 5,000 $\times$ . The scale bars are 20  $\mu\text{m}$  and 2  $\mu\text{m}$ , respectively.



**Figure S3.** SEM micrographs of sample B1 (5 min). Left: 500 $\times$ ; right: 5,000 $\times$ . The scale bars are 20  $\mu\text{m}$  and 2  $\mu\text{m}$ , respectively.

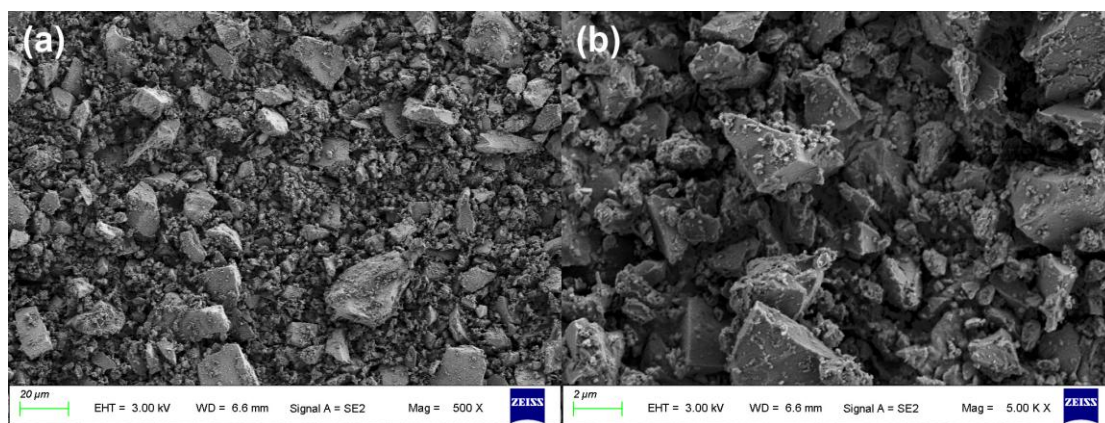


**Figure S4.** SEM micrographs of sample B2 (10 min). Left: 500×; right: 5,000×. The scale bars are 20 μm and 2 μm, respectively.

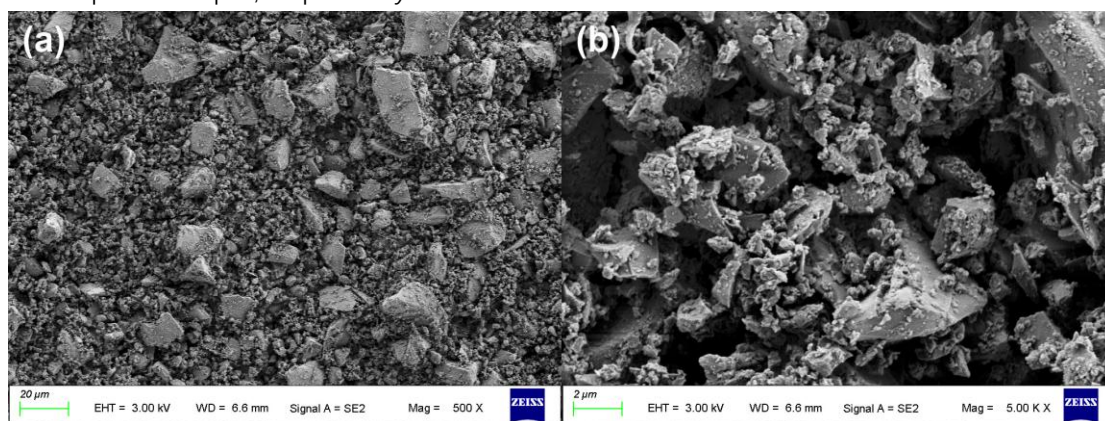


**Figure S5.** SEM micrographs of sample B3 (15 min). Left: 500×; right: 5,000×. The scale bars are 20 μm and 2 μm, respectively.

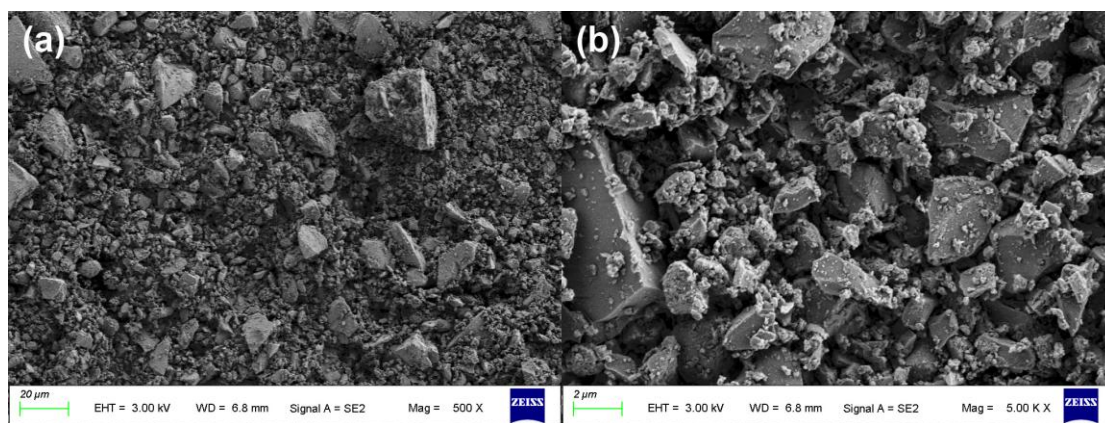




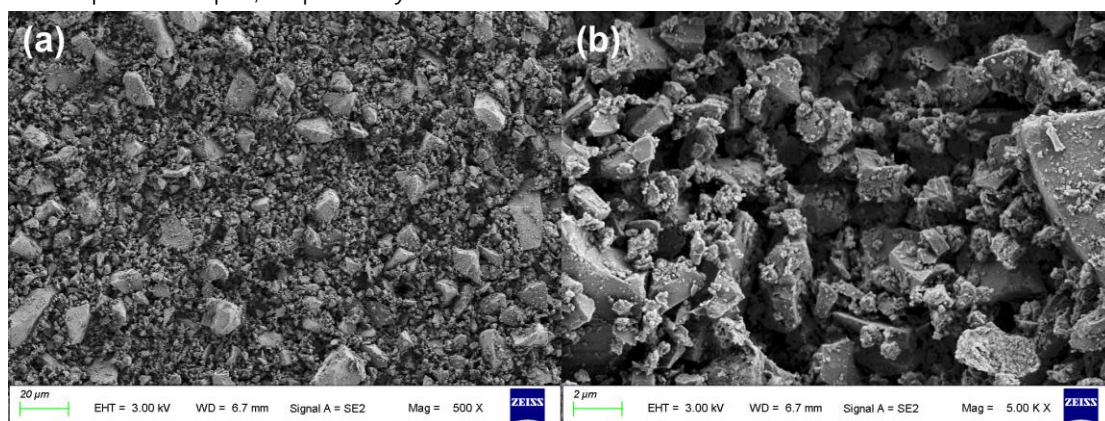
**Figure S6.** SEM micrographs of sample B4 (20 min). Left: 500×; right: 5,000×. The scale bars are 20 μm and 2 μm, respectively.



**Figure S7.** SEM micrographs of sample B5 (25 min). Left: 500×; right: 5,000×. The scale bars are 20 μm and 2 μm, respectively.



**Figure S8.** SEM micrographs of sample B6 (30 min). Left: 500×; right: 5,000×. The scale bars are 20 μm and 2 μm, respectively.



**Figure S9.** SEM micrographs of sample B7 (35 min). Left: 500×; right: 5,000×. The scale bars are 20 μm and 2 μm, respectively.